

The Ethical Hacker's Command Handbook

Chapter 2 – Part 7: Output Formats, Reporting & Saving Scan Results

A scan that isn't saved is a scan you'll probably have to repeat.

Learning Objectives

Learn how to save Nmap results in different formats, choose the right output type, organize scan files, and build good reporting habits.

Why Save Scan Results?

Saving scan results preserves evidence, supports report writing, enables comparisons over time, and allows results to be imported into other tools.

nmap -oN scan.txt

Purpose: Normal output

What it does: Human-readable text output.

When to use: Most manual scans and lab notes.

Avoid when: Machine parsing.

Memory Tip: N = Normal

nmap -oX scan.xml

Purpose: XML output

What it does: Machine-readable XML.

When to use: Automation and tool integration.

Avoid when: Manual reading only.

Memory Tip: X = XML

nmap -oG scan.grep

Purpose: Grepable output

What it does: Easy to filter with grep and scripts.

When to use: Searching large scan results.

Avoid when: General reading.

Memory Tip: G = Grep

nmap -oA office_scan

Purpose: All formats

What it does: Creates .nmap, .xml, and .gnmap files.

When to use: Recommended for most assessments.

Avoid when: Only one format is needed.

Memory Tip: A = All

nmap -v

Purpose: Verbose mode

What it does: Shows additional scan progress.

When to use: Long scans.

Avoid when: Minimal output preferred.

Memory Tip: v = Verbose

nmap -vv

Purpose: Very verbose

What it does: Displays even more detail.

When to use: Troubleshooting.

Avoid when: Simple scans.

Memory Tip: vv = More detail

nmap -d

Purpose: Debug mode

What it does: Shows debugging information.

When to use: Learning and troubleshooting.

Avoid when: Normal assessments.

Memory Tip: d = Debug

nmap --append-output -oN scans.txt

Purpose: Append output

What it does: Adds results to an existing file.

When to use: Maintaining one scan log.

Avoid when: Separate files are preferred.

Memory Tip: Append, don't overwrite

nmap --resume scan.xml

Purpose: Resume scan

What it does: Continues an interrupted scan.

When to use: Long-running scans.

Avoid when: Completed scans.

Memory Tip: Resume = Continue

Comparison Table

Goal	Command
Human-readable	-oN
XML	-oX
Grepable	-oG
All formats	-oA
Verbose	-v / -vv
Debug	-d
Append output	--append-output
Resume scan	--resume

Best Practices

Use descriptive filenames (for example, thm_blue_initial.nmap), avoid overwriting previous scans, and prefer -oA for complete documentation.

Practice Questions

1. What is the difference between -oN, -oX, -oG and -oA?
2. When should XML output be used?
3. What does --append-output do?
4. How does --resume help?
5. Why is -oA commonly recommended?